DEEP REINFORCEMENT LEARNING AND DAMPED LEAST SQUARES INVERSE KINEMATICS METHOD FOR 7-DEGREE-OF-FREEDOM TRAJECTORY TRACKING

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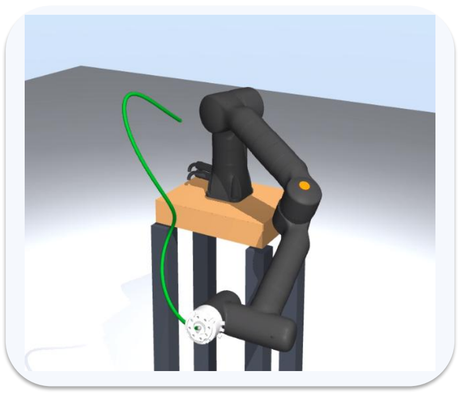
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ABSTRACT

We investigate sequential inverse kinematics for the 7-DOF Kassow KR810. Adaptive damped least squares (DLS) serves as the baseline; Multistep PPO–DLS Inverse Kinematics (MPDIK) introduces policy-guided joint seeds when DLS fails. On 630 MuJoCo trajectories, MPDIK raises complete-trajectory success from 78.3% to 91.9% and reduces position RMSE from 28.26 to 2.01 mm.

INTRODUCTION



KR810 in MuJoCo
(7 revolute joints)

- **The challenge**

  A redundant arm can reach the same pose with multiple joint configurations.

- **Sequential effects**

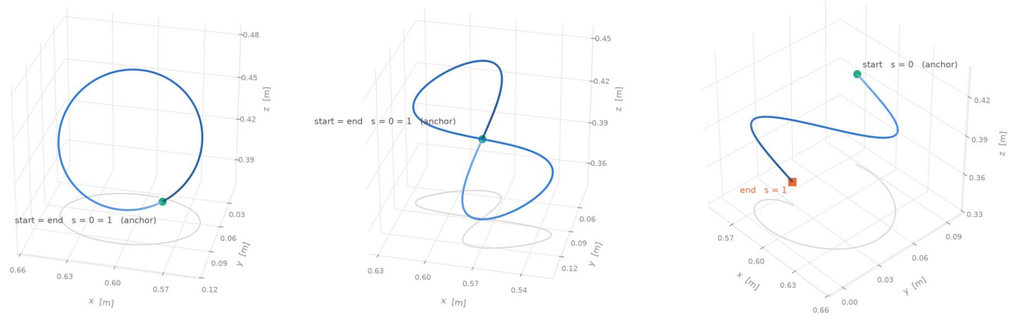
  A poor solution at waypoint k becomes the initial seed for k + 1.

Goal: accurate pose tracking with continuous joint transitions.

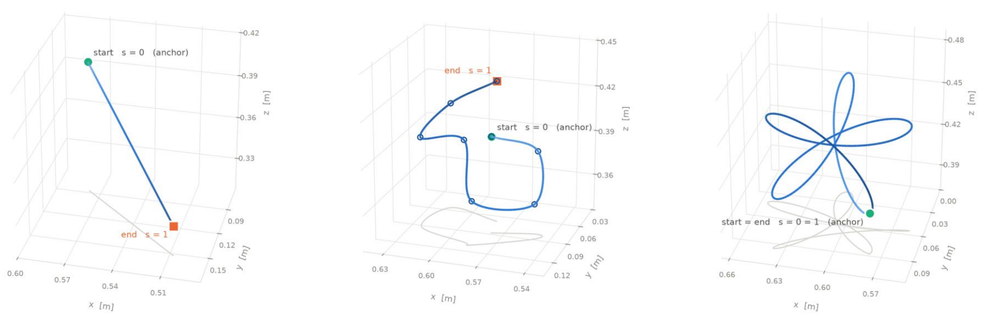
RELATED WORK

Yu and Tan (2025) combined PPO and DLS for individual target poses. This study evaluates the approach on sequences of Cartesian waypoints, where the solution at one waypoint seeds the next.

EXPERIMENTAL BENCHMARK



CIRCLE	FIGURE-EIGHT	HELIX
72 trajectories	72 trajectories	72 trajectories

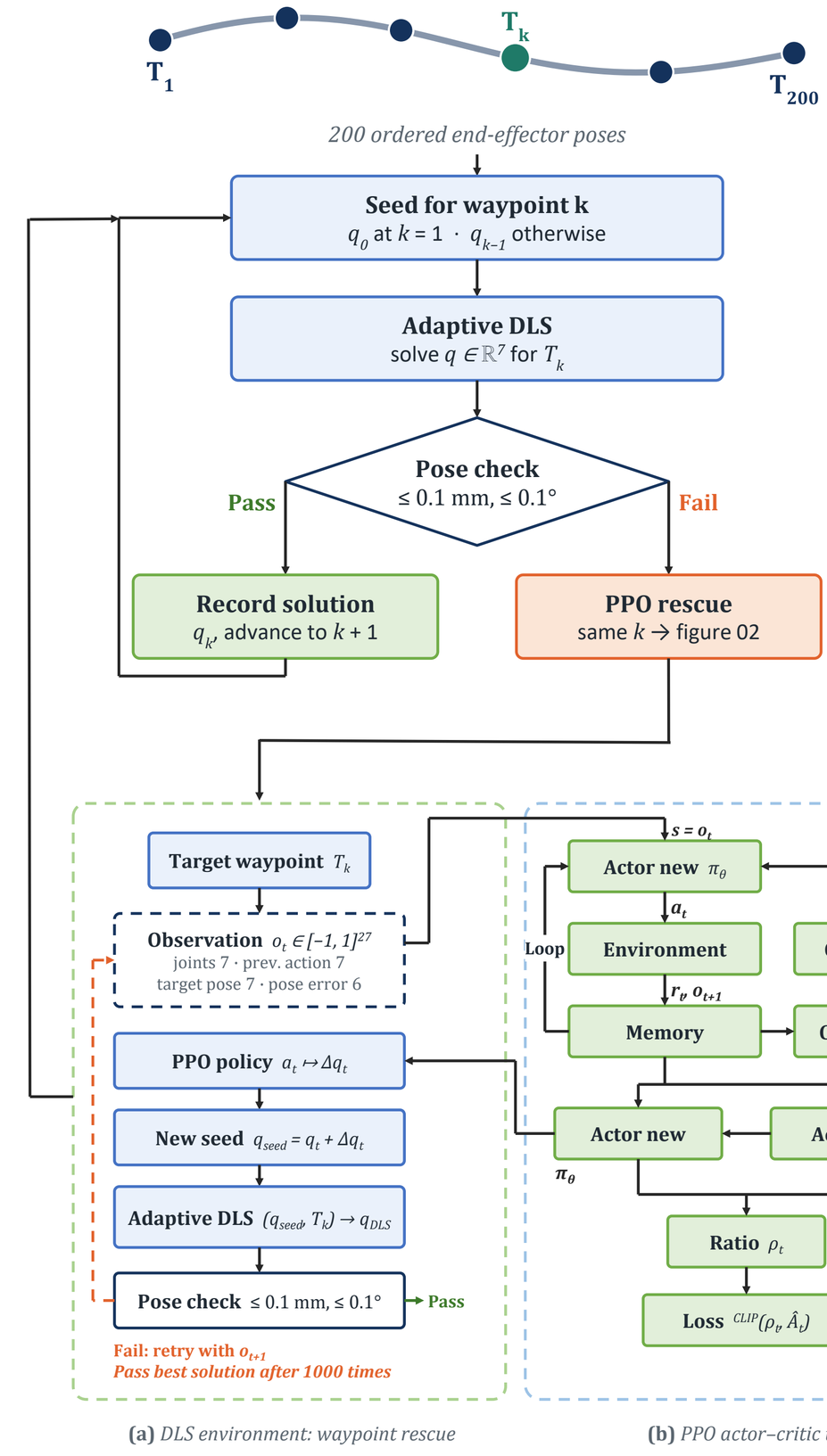


LINE	FREE-FORM	CHALLENGE
72 trajectories	72 trajectories	270 trajectories

Success thresholds: Position error ≤ 0.1 mm and orientation error ≤ 0.1°.

PROPOSED METHOD

MPIDK ALGORITHM



(a) *DLS environment: waypoint rescue* **(b)** *PPO actor–critic training*

EXPERIMENTAL RESULTS

Benchmark results for DLS and MPDIK

Metric	DLS	MPDIK
Successful waypoints	122,909 (97.5%)	125,274 (99.4%)
Completed trajectories	493 (78.3%)	579 (91.9%)
Overall position RMSE	28.26 mm	2.01 mm
Overall orientation RMSE	5.22°	0.33°